

# Research Project - Mathematical Statistics

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## Introduction

For our research project, we have chosen to explore and study some of the most relevant characteristics describing the companies that were part of the S&P 500 in the year 2020.

In this elaborate we investigate the relationship between the Sensitivity Index of those companies and certain relevant attributes listed below.

The Market Sensitivity Index (or simply beta) measures a firm's exposure to overall market trends. We computed it as the ratio between the covariance of the firm's monthly returns in 2020 with market monthly returns in the same year, and the variance of monthly market returns (using log-scaled values).

Formally: Monthly log-returns for firm  $i$  and for the market  $m$  are computed as

$$R_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right), \quad R_{m,t} = \ln\left(\frac{P_{m,t}}{P_{m,t-1}}\right),$$
$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}.$$

where  $P_{i,t}$  and  $P_{m,t}$  denote, respectively, the monthly closing price of firm  $i$  and of the market index at time  $t$  (Logarithmic returns are used because they are time-additive and commonly adopted in empirical asset pricing).

This Index captures the non-diversifiable component of risk and reflects the firm's systematic risk relative to the market.

Our final aim is to understand what drives firms' Market Sensitivity and to assess whether differences in that, across companies, can be explained by firm characteristics and industry sectors, rather than being purely random.

A potential concern in cross-sectional analyzes of Market Sensitivity is that the latter estimates may be distorted by low liquidity, thin trading, or measurement errors, like in the example of smaller firms.

For all those reasons, we decided to focus only on the companies of the S&P 500, which depicts an overall reliable behavior of the U.S. equity market, covering firms that are large, highly liquid, and subject to strict reporting standards mitigating the consequences of these issues.

## Dataset

For our analysis, we used two main datasets to obtain all the information we needed, and initial data preparation was carried out in Excel. Firm-level financial information for S&P 500 companies was obtained from a publicly available dataset hosted on Kaggle.

Monthly stock price data for individual firms and for the S&P 500 index was collected from publicly accessible financial data sources (the website "Stooq") and used to construct a measure of Market Sensitivity.

Specifically, firm-level Sensitivity to market movements was computed using monthly log-returns over the period December 2019 to December 2020. The resulting Sensitivity Index was then merged with the financial dataset using company tickers as unique identifiers, producing a final cross-sectional dataset.

A schematic description of the dataset features, with the abbreviations used in the R script, follows:

- **Symbol** - ticker symbol of a company
- **Name** - complete name of a company
- **Sector** - industry sector classification
- **Price** - latest observed equity price per share (at the dataset reference date)
- **Price.Earnings** - P/E ratio: market price per share divided by earnings per share

- **Dividend.Yield** - dividends per share divided by price
- **Earnings.Share** - earnings per share, net income attributable to common shareholders per share
- **X52.Week.Low** - lowest traded price over the previous 52 weeks
- **X52.Week.High** - highest traded price over the previous 52 weeks
- **Market.Cap** - market capitalization (price x shares outstanding), a measure of firm size
- **EBITDA** — earnings before interest, taxes, depreciation and amortization; operating performance proxy
- **Price.Sales** — price-to-sales ratio (valuation multiple relative to revenues)
- **Price.Book** — price-to-book ratio (valuation multiple relative to accounting equity)
- **Market.Sensibility.Index** — Market Sensitivity measure the firm's exposure to movements in the Market Index

## Data Cleaning

First of all, in order to make the dataset cleaner and more easily manageable, we formatted the data and renamed 'Sensitivity' the 'Market.Sensibility.Index' to make it more readable, leaving the other columns' titles the same since they were already easy to work with and self-explanatory.

Through the `summary()` function in R we witnessed that the values of our dataset covered a wide dispersion in the magnitude of several variables, so we expected the presence of many outliers.

In the context of financial and accounting, data outliers are not necessarily indicative of data errors but instead reflect genuine economic differences across firms.

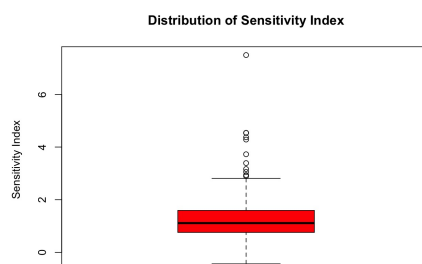
Nevertheless, the outliers may affect the stability of linear regression estimates due to their enormous variability in magnitude. To address this issue and reduce the possibility of errors in the further analyzes, we apply logarithmic transformations to highly skewed variables, reducing asymmetry and the influence of extreme observations.

## Visual Exploration

Before diving into the regression analysis, we conducted a preliminary visual inspection of the dataset aimed to gain some insight on the distribution of the Market Sensitivity Index and its relationship with firm size.

Due to the cross-sectional nature of the data, we focused on three informative graphs: the distribution of the Sensitivity Index through a box plot, its relationship with EBITDA by a scatter plot, and its relationship with Market Capitalization, also with a scatter plot. This exploratory step should be interpreted as descriptive rather than conclusive, since bi-variate plots do not control for correlations among regressors; note also that the EBITDA and the Market Capitalization are strongly correlated between them (as we can see via the correlation matrix).

The box plot depicts the range of values attained by the Sensitivity Index of the different firms. It is easily observable that its values are concentrated close to one (consistent with our interpretation of beta), although also presenting a non-negligible number of outliers, which are not unexpected in firm-level finance data as stated before. This is due to the fact that companies can differ substantially on many aspects, such as leverage, cyclicity, business risk, etc.

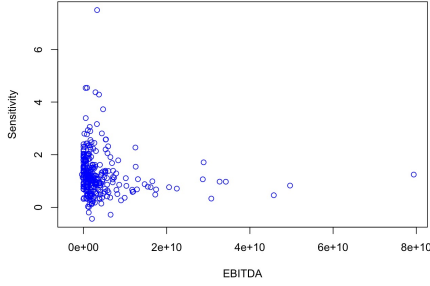


Proceeding with the scatter plot of Sensitivity against EBITDA, there emerge two main aspects. It is immediately visible that the EBITDA is highly right-skewed, with most observations concentrated at low values.

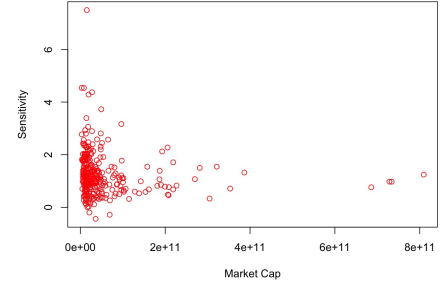
Furthermore, the variance of the Sensitivity Index is not constant, large for firms with smaller EBITDA and lower for larger companies (which means, generally, greater EBITDA), indicating heteroscedasticity with respect to operating scale.

A similar structure emerges when plotting Sensitivity against Market Capitalization. Market Cap spans multiple orders of magnitude, producing a strong concentration of points near the origin. As before, the variance of beta appears to be higher for smaller-cap firms and lower for the largest companies. Importantly, visual evidence does not suggest a clear linear trend in levels; instead, the dominant feature is the scale imbalance in the explanatory variable.

For these reasons, the analysis proceeds with a logarithmic transformation of Market Capitalization and of EBITDA (when positive), to reduce skewness, heteroscedasticity and improve the stability and interpretability of subsequent regression estimates.

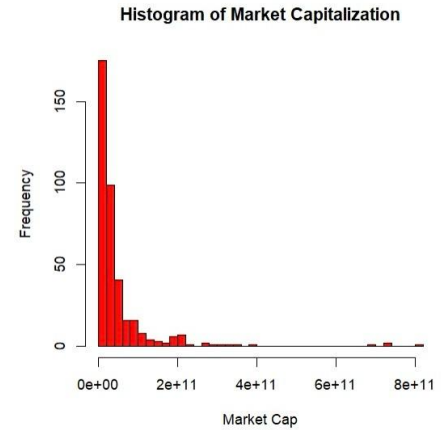
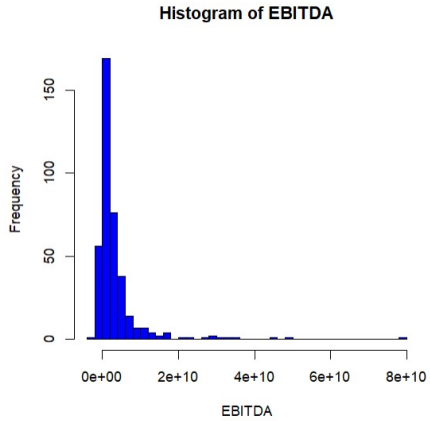


(a) Sensitivity vs EBITDA



(b) Sensitivity vs Market Cap

We concluded the first visual analysis with both histograms showing a strong right-skewed distribution, with most firms concentrated at relatively low capitalization and EBITDA levels and a small number of very large firms forming a long right tail. The logarithmic transformation will help the asymmetry also in this case as stated before.



## Multiple Linear Regression

Following the visual analysis, we can proceed with the discussion of the regression.

Firstly we transformed ‘Sector’ into a categorical variable and we set Health Care as the “baseline”. The latter choice finds its justification in “Health Care” being widely regarded as one of the most stable industries with respect to macroeconomic fluctuations.

Demand for health care services is largely inelastic as it is primarily driven by demographic factors and regulatory frameworks. As a result, firms in this sector tend to exhibit relatively moderate and stable exposure to market movements, making it a natural benchmark against which to compare more industries. By doing this, each sector coefficient in the regression measures the difference in average Market Sensitivity between that sector and Health Care, holding firm size and valuation constant.

For our purposes, we run a multiple linear regression with the Sensitivity Index as the dependent variable and sectorial affiliation, firm size (logarithm of Market Capitalization, logarithm of EBITDA), and valuation

multiples (Price-to-Earnings, Price-to-Book, and Price-to-Sales ratios) as predictors.

This specification allows us to disentangle structural industry effects from cross-sectional differences in firm characteristics, providing an explanatory framework rather than a predictive one. This baseline model is intentionally broad and aims to capture potential determinants of systematic risk while controlling for size, profitability, and valuation effects.

The results reveal substantial sectoral heterogeneity in beta values, with cyclical industries exhibiting higher Sensitivity and defensive sectors displaying lower exposure to market movements. In contrast, several firm-level controls are not statistically significant once sector affiliation is taken into account, suggesting potential redundancy among regressors, motivating thus a subsequent model selection procedure. Overall, the baseline regression confirms that both industry affiliation and firm-specific characteristics contribute to explaining cross-sectional variation in systematic risk.

The significance of the regression model is confirmed by the F-test, which strongly rejects the null hypothesis that all slope coefficients are jointly equal to zero ( $p\text{-value} < 2.2 \times 10^{-16}$ ). Moreover, the model explains a non-trivial proportion of the cross-sectional variation in the Sensitivity Index, with an adjusted  $R^2$  of approximately 0.386. Among the valuation measures considered, the Price-to-Earnings ratio emerges as the only robustly significant predictor, exhibiting a negative coefficient. This result indicates that within the sample, firms with higher P/E ratios tend to display lower estimated Market Sensitivity.

## BIC-based model selection

Given the presence of correlated regressors and the potential for redundant explanatory variables, we subsequently employ the Bayesian Information Criterion (BIC) to guide model selection.

BIC penalizes model complexity more strongly than alternative criteria, favoring parsimonious specifications that balance explanatory power with interpretability.

The BIC-selected model retains sectorial indicators, Market Capitalization, and the P/E ratio as the key determinants of Market Sensitivity, while excluding additional profitability and valuation measures that do not provide incremental explanatory power.

The final regression confirms that sector affiliation remains a dominant driver of beta, even after controlling for firm characteristics, underscoring the structural nature of industry-level exposure to macroeconomic risk.

In addition, the negative association between Market Capitalization and Sensitivity reinforces the role of diversification, while the inverse relationship between P/E ratios and beta suggests that firms perceived by the market as more stable and less cyclical exhibit a lower exposure to systematic fluctuations.

## ANOVA

An Analysis of Variance (ANOVA) is employed to test for statistically significant differences in the mean Sensitivity Index across industry sectors. The analysis evaluates whether sectoral affiliation explains systematic variation in Market Sensitivity beyond firm-level heterogeneity.

We first estimate a one-way ANOVA with sector affiliation as the sole explanatory factor, testing the null hypothesis that average Market Sensitivity is equal across sectors. This hypothesis is strongly rejected, indicating substantial heterogeneity in systematic risk across industries. Such heterogeneity is consistent with economic intuition, as firms operate in different macroeconomic environments and are exposed to varying degrees of cyclical, regulatory, and external risk.

We then estimate a conditional ANOVA by including firm size and valuation as additional controls, in order to assess whether sectoral differences persist after accounting for observable firm characteristics. Sector affiliation remains highly statistically significant in this specification, confirming that industry effects explain a meaningful portion of cross-sectional variation in Market Sensitivity beyond size and valuation effects.

These findings indicate that systematic risk is shaped not only by firm-level attributes but also by structural, sector-specific factors related to economic exposure and macroeconomic conditions.

## Tukey Post-Hoc Test

Following the ANOVA results, we performed a Tukey Honestly Significant Difference (HSD) post-hoc test to identify which specific pairs of sectors differ in terms of average Market Sensitivity. The results reveal economically meaningful and statistically significant differences among the different sectors.

In particular, the Energy field shows a significantly higher Market Sensitivity compared to the other sectors, confirming its strong exposure to macroeconomic and external factors.

Conversely, other sectors such as Utilities and Consumer Staples tend to exhibit a more defensive attitude, with significantly lower beta levels relative to more cyclical sectors.

## Conclusions

Exploratory analysis showed that beta is generally centered around values close to one, indicating that most firms broadly track market movements, while a non-negligible number exhibit unusually high or low Sensitivity. Scatterplots against Market Capitalization and EBITDA revealed strong scale imbalances and greater dispersion among smaller firms, motivating the use of logarithmic transformations for size-related variables.

A baseline multiple linear regression confirms that sector affiliation plays a central role in explaining cross-sectional variation in beta, while firm-level characteristics provide more limited explanatory power once sector effects are included. Among valuation measures, the P/E ratio emerges as the only robustly significant predictor, displaying a negative association with Market Sensitivity. BIC-based model selection further supports a parsimonious specification retaining sector affiliation, Market Capitalization, and Price-to-Earnings as the key determinants.

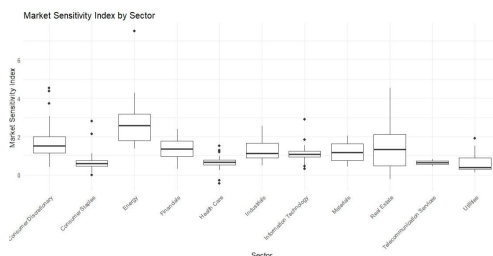
ANOVA results provide strong evidence that average beta differs significantly across sectors, both unconditionally and after controlling for firm size and valuation. Post-hoc Tukey tests reveal economically meaningful patterns: cyclical sectors such as Energy exhibit higher Sensitivity, while defensive sectors including Utilities and Consumer Staples display significantly lower beta values.

Overall, the findings suggest that systematic risk among large U.S. firms is primarily shaped by industry-level characteristics, with firm size and valuation providing secondary but meaningful contributions once sector heterogeneity is taken into account.

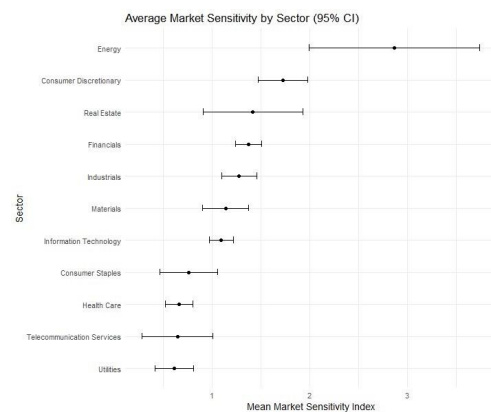
This study, though, is subject to several limitations. First, the analysis is based on a single cross-sectional snapshot, with Market Sensitivity estimated using monthly returns over the year 2020. Given the exceptional economic conditions during this period, including heightened market volatility related to the COVID-19 pandemic, the estimated beta values may not be fully representative of firms' long-run systematic risk.

Second, while the model includes key firm-level and sectorial controls, other relevant determinants of beta—such as financial leverage, geographic exposure, or business segment diversification—are not explicitly incorporated, which may lead to omitted variable bias.

Finally, the use of accounting and valuation measures introduces potential measurement error, and the reliance on publicly available data limits the scope of firm-specific information that can be exploited.



(a) Market Sensitivity Index by Sector



(b) Average Beta by Sector (95% CI)

# APPENDIX

## Bibliography

- Stooq: <https://stooq.com>
- Kaggle S&P 500 Dataset: <https://www.kaggle.com/datasets/paytonfisher/sp-500-companies-with-financial-information>
- U.S. Securities and Exchange Commission (SEC): <https://www.sec.gov>

```
Call:
lm(formula = Sensitivity ~ Sector + logmarket + log_EBITDA +
    Price.Book + Price.Earnings + Price.Sales, data = data)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.5883 -0.3568 -0.0921  0.2313  3.7759
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.832e+00  1.147e+00   4.212 3.61e-05 ***
SectorConsumer Staples -9.394e-01  2.039e-01  -4.607 6.70e-06 ***
SectorEnergy    1.082e+00  2.233e-01  4.847 2.29e-06 ***
SectorFinancials -6.270e-01  2.308e-01  -2.717 0.00707 **
SectorHealth Care -9.325e-01  1.806e-01  -5.164 5.17e-07 ***
SectorIndustrials -5.105e-01  1.604e-01  -3.183 0.00166 **
SectorInformation Technology -4.279e-01  1.649e-01  -2.595 0.01007 *
SectorMaterials  -6.562e-01  2.012e-01  -3.261 0.00127 **
SectorReal Estate -1.573e-01  2.441e-01  -0.645 0.51986
SectorTelecommunication Services -8.590e-01  5.262e-01  -1.633 0.10390
SectorUtilities  -1.163e+00  1.992e-01  -5.837 1.76e-08 ***
logmarket     -9.433e-02  1.256e-01  -0.751 0.45343
log_EBITDA    -3.210e-02  1.175e-01  -0.273 0.78499
Price.Book     3.541e-06  5.279e-04  0.007 0.99465
Price.Earnings -3.226e-03  1.074e-03  -3.003 0.00296 **
Price.Sales    -2.536e-02  1.897e-02  -1.337 0.18268
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.6994 on 234 degrees of freedom
(139 osservazioni eliminate a causa di valori mancanti)
Multiple R-squared:  0.4228,    Adjusted R-squared:  0.3858
F-statistic: 11.43 on 15 and 234 DF,  p-value: < 2.2e-16
```

(a) non optimal regression

```
Call:
lm(formula = Sensitivity ~ Sector + logmarket + Price.Earnings,
    data = data_wo_NA)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.6344 -0.3610 -0.1047  0.2437  3.6679
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.0960555  1.1091517   4.595 7.05e-06 ***
SectorConsumer Staples -0.9331449  0.1977873  -4.718 4.07e-06 ***
SectorEnergy    1.0626438  0.2222446  4.781 3.06e-06 ***
SectorFinancials -0.7211000  0.2206091  -3.269 0.001241 **
SectorHealth Care -0.9970358  0.1713121  -5.820 1.90e-08 ***
SectorIndustrials -0.5233826  0.1597637  -3.276 0.001211 **
SectorInformation Technology -0.5051548  0.1546721  -3.266 0.001252 **
SectorMaterials  -0.6795746  0.2000064  -3.398 0.000797 ***
SectorReal Estate -0.3615528  0.1955559  -1.849 0.065725 .
SectorTelecommunication Services -0.8528778  0.5164593  -1.651 0.099983 .
SectorUtilities  -1.2036153  0.1829417  -6.579 3.00e-10 ***
logmarket     -0.1362810  0.0464976  -2.931 0.003710 **
Price.Earnings -0.0034294  0.0009349  -3.668 0.000302 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.6979 on 237 degrees of freedom
Multiple R-squared:  0.4177,    Adjusted R-squared:  0.3882
F-statistic: 14.17 on 12 and 237 DF,  p-value: < 2.2e-16
```

(b) optimal regression

```
> ##Now we run an ANOVA test to understand if sector affiliation is meaningful or not##
> ANOVA_1 = aov(Sensitivity ~ Sector, data = data_wo_NA)
```

```
> summary(ANOVA_1)
              Df Sum Sq Mean Sq F value Pr(>F)
Sector         10   70.2    7.020   13.1 <2e-16 ***
Residuals     239  128.1    0.536
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
> ANOVA_2 = aov(Sensitivity ~ Sector + logmarket + Price.Earnings, data = data_wo_NA)
```

```
> summary(ANOVA_2)
              Df Sum Sq Mean Sq F value    Pr(>F)
Sector         10   70.20    7.020   14.41 < 2e-16 ***
logmarket       1    6.07    6.068   12.46 0.000500 ***
Price.Earnings  1    6.55    6.534   13.46 0.000302 ***
Residuals     237  115.45    0.487
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(a) ANOVA

```
> ##Correlation Matrix##
```

```
> cor(X)
           log_EBITDA logmarket Price.Earnings Price.Book Price.Sales
log_EBITDA  1.00000000  0.8223055  -0.13528388  0.08478653 -0.15542475
logmarket   0.82230552  1.0000000   0.16438209  0.11563028  0.16126535
Price.Earnings -0.13528388  0.1643821  1.00000000  0.02339009  0.24288618
Price.Book    0.08478653  0.1156303  0.02339009  1.00000000  0.01158444
Price.Sales   -0.15542475  0.1612653  0.24288618  0.01158444  1.00000000
```

(b) Correlation Matrix